ID72001 Assessment 3 — Starter Repository

A secure, observable microservice platform on Kubernetes, orchestrated with Terraform, on **AWS, Azure or Catalyst Cloud** — you choose.

This repository is **deliberately incomplete**. About half of it is supplied so you are not retyping lab code; the rest is yours to write, and the assessment measures that half.

> **Start with `TASKS.md`.** It lists every file as *given*, *worked example* or *to write*, says which task and how many marks each one carries, and explains why the line falls where it does.

```
./scripts/check-progress.sh          # what is still unwritten
./scripts/check-progress.sh azure    # once you have chosen a cloud
```

What you are building

Component	Role	Reachable from outside?
`frontend`	Renders the page. Owns no data.	Yes — via the Ingress, over HTTPS
`catalog`	Owns the product list. No dependencies.	No
`orders`	Validates orders, writes to Redis.	No
`redis`	Order store, on a PersistentVolume	No

Published on your own hostname, served over TLS from a certificate authority you build inside the cluster, and monitored end to end by Prometheus across five layers.

One Ingress reaches the internet. Everything else is `ClusterIP`.

Where to read what

If you want to…	Read
Know what to write and what you were given	`TASKS.md`
Know what stage 1 must produce, on any cloud	`docs/CLOUD-CONTRACT.md`
Know the order to do things in	`RUNBOOK.md`
Know what a specific file must contain	the skeleton file itself — each one carries its own requirements, traps and checks

If you want to…	Read
Understand Ingress and cert-manager — neither was in the labs	`docs/INGRESS-AND-TLS-PRIMER.md`, then `docs/examples/EXAMPLE-ingress.yaml`
Fix something broken	`docs/TROUBLESHOOTING.md`
Write Task 8	`docs/CRITICAL-EVALUATION-TEMPLATE.md`
Know how the dashboard is marked	`monitoring/README-dashboard.md`

The skeletons are where the detail lives. `manifests/06-tls.yaml` is a hundred lines of comments and no YAML; that is not an oversight.

Choosing a cloud

All three are equally creditable and Criterion 1 is marked identically across them. Only stage 1 differs — roughly 85% of this repository is the same whichever you pick, including the bootstrap script.

- **AWS Academy Learner Lab** — credentials expire every few hours, which interrupts long applies; no managed DNS, so you use `sslip.io`.
- **Azure** — the school subscription is stable and you get a free hostname. Watch the ~4 vCPU cap, the restricted VM sizes and the per-student allowed regions.
- **Catalyst Cloud** — a New Zealand provider, so data stays onshore. Plain OpenStack, which makes it the most transferable of the three.

`docs/CLOUD-CONTRACT.md` has the full comparison. `RUNBOOK.md` §0 has the pre-flight commands for each — run them before you commit to one, because quota, not skill, is what usually decides this.

Delete the two `terraform/infra-*` directories you do not use before you submit.

How much comes from the labs

Task	Source
1 — Terraform, cloud foundation	New. Lab structure carries over; the provider does not
2 — Platform orchestration	New, and cloud-neutral
3 — Microservices and images	**Lab 9.3**, nearly unchanged — the source is supplied
4 — Workloads, config, storage	**Lab 9.3** manifests, plus ConfigMap/Secret from **9.2** and a PVC
5 — Hostname and TLS	New, and the hardest thing here
6 — Prometheus, end to end	**Lab 9.4** — instrumentation, ServiceMonitors, rules
7 — Grafana dashboard	**Lab 9.4** — same panel ideas, your namespace
8–9 — Evaluation and defense	Entirely your own

Go back to your own lab work before you go to the internet. You have built most of this before.

Things that will cost you marks, collected in one place

Clicking instead of coding. Resources in Tasks 1 and 2 created by hand in a web console earn nothing in those criteria, even if the system works.

A committed secret. A live credential, private key, kubeconfig or real password anywhere in the submitted archive caps Criterion 4 at *Not achieved*. `.gitignore` already excludes state, `.terraform/` and `k8s-key*` — do not weaken it.

Storage the platform can discard. On Azure especially, `/mnt` is a temporary disk whose contents are destroyed on deallocation. The supplied bootstrap mounts your data disk at `/datadisk` on every cloud for exactly this reason. Check with `findmnt /datadisk` rather than assuming.

A hardcoded hostname. `scripts/render.sh` substitutes `__SHOP_HOST__`. Type your own hostname into the manifests instead and the repository stops being reproducible for anyone, including your marker.

One self-signed certificate instead of a chain. If subject and issuer match, Criterion 5 caps at a bare pass however well the site works.

A dashboard that only exists in your Grafana. Export the JSON and commit it. The VM is gone by the time it is marked.

A working deployment you cannot explain. Tasks 8 and 9 are 20 marks for reasoning, not building. Every criterion's upper bands ask *why*, not *does it run*.

Platform-independent gotchas worth knowing now

- **The ServiceMonitor selector.** kube-prometheus-stack only discovers ServiceMonitors and rules carrying its release label unless you tell it otherwise. Yours will be created successfully and silently ignored. Nothing is logged. See `monitoring/servicemonitors.yaml`.
- `flask_http_request_total` **has no** `endpoint` **label.** Group a counter by a label it does not carry and PromQL returns one empty-labelled series rather than an error. See `monitoring/alerts.yaml`.
- **One gunicorn worker.** `prometheus_client` holds counters in process memory, so several workers each report a partial count. Scale by Pod.
- **ReadWriteOnce and rolling updates do not mix.** See `manifests/02-redis.yaml`.
- **Two Terraform stages, not one.** Provider configuration is evaluated at plan time, before the cluster exists. See `terraform/platform/main.tf`.

Each of these is a question you may be asked in the code defense.